Evidence-Weighted Policy Gradient Convergence in General Stochastic Games

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Abstract

We introduce an *evidence-weighted* variant of the policy gradient method for multi-agent learning in general stochastic games. Building on the convergence framework of Giannou, Lotidis, Mertikopoulos, and Vlatakis-Gkaragkounis (2022), we show that when agents scale their gradient updates by a Keynesian weight of evidence V_i, the effective variance in the Lyapunov convergence analysis improves by a factor of $\mathrm{HM}(V)/\,\mathrm{AM}(V) \leq 1$, where HM and AM denote the harmonic and arithmetic means of the evidence weights across agents. The convergence rate exponent is preserved at $\mathcal{O}(1/\sqrt{n})$ for REINFORCE, but the leading constant is strictly tighter whenever agents have heterogeneous evidence quality. The evidence-weighting scheme arises naturally from the Keynesian loss function in the Ω-framework, connecting multi-agent reinforcement learning to the epistemology of incomplete knowledge.

Contents

1 Setup and Notation **2**

2 Standard Policy Gradient and Giannou's Results **2**
2.1 The PG template . . . 2
2.2 Giannou's convergence results . . . 3

3 Evidence-Weighted Policy Gradient **3**
3.1 Motivation from the Ω-framework . . . 3
3.2 The algorithm . . . 3

4 Modified Lyapunov Analysis **4**
4.1 The modified energy inequality . . . 4
4.2 Variance scaling . . . 4

5 The AM-HM Variance Improvement **4**
5.1 The Keynesian variance model . . . 5
5.2 Unweighted vs. weighted total variance . . . 5
5.3 The main result . . . 5

6 Convergence of Evidence-Weighted PG **6**

7 Optimal Evidence Weight **7**

8 Connection to the Ω-Framework **7**

9 Discussion **8**

A Pseudocode **9**

1 Setup and Notation

We adopt the stochastic game framework of Giannou et al. [1]. An N-player stochastic game $\mathcal{G} = (\mathcal{S}, \mathcal{N}, \{\mathcal{A}_i\}_{i\in\mathcal{N}}, P, \zeta, \rho)$ consists of:

- A finite set of agents $\mathcal{N} = \{1, \dots, N\}$ and states $\mathcal{S} = \{1, \dots, S\}$.
- For each agent i, a finite action space $\mathcal{A}_i$ with $|\mathcal{A}_i| = A_i$.
- Reward functions $R_i : \mathcal{S} \times \mathcal{A} \to [-1, 1]$.
- Markov transition kernel $P(\cdot \mid s, \alpha)$ and termination probabilities $\zeta_{s,\alpha} > 0$.
- Initial state distribution $\rho \in \Delta(\mathcal{S})$.

A *stationary Markovian policy* for agent i is a map $\pi_i : \mathcal{S} \to \Delta(\mathcal{A}_i)$. We write $\Pi_i := \Delta(\mathcal{A}_i)^S$ for agent i's policy space, $\Pi := \prod_i \Pi_i$ for the joint policy space, and $\pi = (\pi_i)_{i\in\mathcal{N}} \in \Pi$ for a joint policy profile.

The value function of agent i under profile π starting from state s is:

$$V_{i,s}(\pi) := \mathbb{E}_{\tau\sim\mathrm{MDP}}\left[\sum_{t=0}^{T(\tau)} R_i(s_t, \alpha_t) \;\middle|\; s_0 = s\right]. \tag{1}$$

We write $v_i(\pi) = \nabla_i V_{i,\rho}(\pi)$ for the individual policy gradient and $v(\pi) = (v_i(\pi))_{i\in\mathcal{N}}$ for the ensemble.

Definition 1.1 (Nash equilibrium, stability, SOS — Giannou et al. Def. 1–2)**.** *A policy* $\pi^* = (\pi_i^*)_{i\in\mathcal{N}} \in \Pi$ *is:*

(a) Nash *if* $V_{i,\rho}(\pi_i^*; \pi_{-i}^*) \geq V_{i,\rho}(\pi_i; \pi_{-i}^*)$ *for all* i *and all* $\pi_i \in \Delta(\mathcal{A}_i)^S$.

(b) Stable *if* $\langle v(\pi), \pi - \pi^*\rangle < 0$ *for all* $\pi \neq \pi^*$ *sufficiently close.*

(c) Second-order stationary (SOS) *if there exists* $\mu > 0$ *such that*

$$\langle v(\pi), \pi - \pi^*\rangle \leq -\mu\|\pi - \pi^*\|^2 \quad \textit{for all } \pi \textit{ sufficiently close to } \pi^*. \tag{2}$$

2 Standard Policy Gradient and Giannou's Results

2.1 The PG template

At each episode $n = 1, 2, \dots$, agents update their joint policy via:

$$\pi_{n+1} = \mathrm{proj}_\Pi(\pi_n + \gamma_n \hat{v}_n), \tag{PG}$$

where $\gamma_n > 0$ is the step size and $\hat{v}_n$ is a gradient estimate. The signal decomposes as:

$$\hat{v}_n = v(\pi_n) + U_n + b_n, \tag{3}$$

with $U_n = \hat{v}_n - \mathbb{E}[\hat{v}_n \mid \mathcal{F}_n]$ (zero-mean noise) and $b_n = \mathbb{E}[\hat{v}_n \mid \mathcal{F}_n] - v(\pi_n)$ (bias). These satisfy:

$$\mathbb{E}[\|b_n\| \mid \mathcal{F}_n] \leq B_n, \qquad \mathbb{E}[\|U_n\|^2 \mid \mathcal{F}_n] \leq \sigma_n^2, \tag{4}$$

with $B_n = \mathcal{O}(1/n^{\ell_b})$ and $\sigma_n^2 = \mathcal{O}(n^{\ell_\sigma})$.

2.2 Giannou's convergence results

Theorem 2.1 (Giannou et al. Theorem 1 — Convergence to stable Nash)**.** *Let π^* be a stable Nash policy and $\gamma_n = \gamma/(n+m)^p$ with $p \in (1/2, 1]$, $p + \ell_b > 1$, $p - \ell_\sigma > 1/2$. Then for any $\delta > 0$, there exists a neighbourhood $\mathcal{U}$ of π^* such that*

$$\mathbb{P}(\pi_n \to \pi^* \mid \pi_1 \in \mathcal{U}) \geq 1 - \delta,$$

provided γ is small enough relative to δ.

Theorem 2.2 (Giannou et al. Theorem 2 — Rate for SOS Nash)**.** *Under the same hypotheses with π^* SOS (parameter $\mu > 0$), the convergence rate is:*

$$\mathbb{E}\big[\|\pi_n - \pi^*\|^2 \mid \mathcal{E}\big] = \begin{cases} \mathcal{O}(1/n^{2\mu\gamma}) & \text{if } p = 1 \text{ and } 2\mu\gamma < q, \\ \mathcal{O}(1/n^q) & \text{otherwise,} \end{cases}$$

where $q = \min\{\ell_b, p - 2\ell_\sigma\}$. For REINFORCE (Model 3) with $p = 1$, this gives $\mathcal{O}(1/\sqrt{n})$.

The proof rests on the *Lyapunov function*:

$$D_n := \frac{1}{2}\|\pi_n - \pi^*\|^2, \tag{5}$$

satisfying the *energy inequality* (Giannou Lemma A.1):

$$\boxed{D_{n+1} \leq D_n + \gamma_n \langle v(\pi_n), \pi_n - \pi^* \rangle + \gamma_n \xi_n + \gamma_n \chi_n + \gamma_n^2 \psi_n^2,} \tag{6}$$

where $\xi_n = \langle U_n, \pi_n - \pi^* \rangle$, $\chi_n = \|\Pi\| B_n$, and $\psi_n^2 = \frac{1}{2}\|\hat{v}_n\|^2$.

3 Evidence-Weighted Policy Gradient

3.1 Motivation from the Ω-framework

In the Ω-framework (theos, items 30–31), each agent i maintains a *Keynesian weight of evidence* $V_i(\mathcal{X})$ measuring how much data supports its predictions about observation $\mathcal{X} \in O$. The *Keynesian loss* augments the standard prediction loss with a fragility penalty:

$$\mathcal{L}_K(Y, \hat{Y}, V) = \underbrace{\mathcal{L}(Y, \hat{Y})}_{\text{prediction error}} + \mu \cdot \underbrace{V^{-1}}_{\text{fragility penalty}}, \tag{7}$$

whose gradient decomposes as (theos eq. 11):

$$\nabla_\theta \mathcal{L}_K = \underbrace{\nabla_\theta \mathcal{L}(Y, \hat{Y})}_{\text{prediction improvement}} - \underbrace{\mu V^{-2} \nabla_\theta V}_{\text{evidence seeking}}. \tag{8}$$

The fragility penalty V^{-1} implies that agents with *low evidence should be conservative*: they should take smaller steps to avoid acting on unreliable gradient estimates.

3.2 The algorithm

Definition 3.1 (Evidence-weighted PG)**.** *Let $w_{i,n} \in [w_{\min}, 1]$ be an $\mathcal{F}_n$-measurable* evidence weight *for agent i at episode n, with $w_{\min} > 0$. The* evidence-weighted policy gradient *(EWPG) update is:*

$$\pi_{i,n+1} = \mathrm{proj}_{\Pi_i}\big(\pi_{i,n} + \gamma_n \cdot w_{i,n} \cdot \hat{v}_{i,n}\big) \quad \text{for each } i \in \mathcal{N}. \tag{EWPG}$$

In the joint policy space, writing $W_n = \mathrm{diag}(w_{1,n} I_{A_1 S}, \ldots, w_{N,n} I_{A_N S})$, this becomes:

$$\pi_{n+1} = \mathrm{proj}_\Pi(\pi_n + \gamma_n W_n \hat{v}_n). \tag{9}$$

Remark 3.2 (Choice of evidence weight)**.** *The natural choice from the Keynesian framework is $w_{i,n} = V_{i,n}/\max_j V_{j,n}$, ensuring $w_{i,n} \in (0, 1]$. When all agents have equal evidence, $w_{i,n} = 1$ for all i and EWPG reduces to standard PG. We discuss the optimal choice in Section 7.*

4 Modified Lyapunov Analysis

4.1 The modified energy inequality

Lemma 4.1 (Modified energy inequality)**.** *Under EWPG* (EWPG)*, the Lyapunov function* (5) *satisfies:*

$$\boxed{D_{n+1} \leq D_n + \gamma_n \langle W_n v(\pi_n), \pi_n - \pi^* \rangle + \gamma_n \xi_n^w + \gamma_n \chi_n^w + \gamma_n^2 (\psi_n^w)^2,} \tag{10}$$

where:

$$\xi_n^w = \langle W_n U_n, \pi_n - \pi^* \rangle, \tag{11}$$

$$\chi_n^w = \|\Pi\| \cdot \|W_n\| \cdot B_n, \tag{12}$$

$$(\psi_n^w)^2 = \tfrac{1}{2} \|W_n \hat{v}_n\|^2. \tag{13}$$

Proof. By non-expansiveness of the projection proj_Π:

$$\begin{aligned} D_{n+1} &= \frac{1}{2}\|\pi_{n+1} - \pi^*\|^2 = \frac{1}{2}\|\,\mathrm{proj}_\Pi(\pi_n + \gamma_n W_n \hat{v}_n) - \mathrm{proj}_\Pi(\pi^*)\|^2 \\ &\leq \frac{1}{2}\|\pi_n + \gamma_n W_n \hat{v}_n - \pi^*\|^2 \\ &= \frac{1}{2}\|\pi_n - \pi^*\|^2 + \gamma_n \langle W_n \hat{v}_n, \pi_n - \pi^* \rangle + \frac{1}{2}\gamma_n^2 \|W_n \hat{v}_n\|^2 \\ &= D_n + \gamma_n \langle W_n (v(\pi_n) + U_n + b_n), \pi_n - \pi^* \rangle + \frac{1}{2}\gamma_n^2 \|W_n \hat{v}_n\|^2. \end{aligned} \tag{14}$$

Expanding the inner product and bounding $|\langle W_n b_n, \pi_n - \pi^* \rangle| \leq \|W_n\| \cdot \|b_n\| \cdot \|\pi_n - \pi^*\| \leq \|\Pi\| \|W_n\| B_n$ by Cauchy–Schwarz gives the result. □

4.2 Variance scaling

The key observation is how the noise term scales under evidence weighting.

Lemma 4.2 (Variance scaling)**.** *For the weighted noise $W_n U_n$, the conditional variance satisfies:*

$$\mathbb{E}\big[\|W_n U_n\|^2 \mid \mathcal{F}_n\big] = \sum_{i=1}^N w_{i,n}^2 \cdot \mathbb{E}\big[\|U_{i,n}\|^2 \mid \mathcal{F}_n\big] \leq \sum_{i=1}^N w_{i,n}^2 \cdot \sigma_{i,n}^2. \tag{15}$$

Proof. Since $W_n = \mathrm{diag}(w_{1,n}, \ldots, w_{N,n})$ acts on the block-diagonal structure of the joint policy space, and $U_{i,n}$ are the per-agent noise components:

$$\|W_n U_n\|^2 = \sum_{i=1}^N \|w_{i,n} U_{i,n}\|^2 = \sum_{i=1}^N w_{i,n}^2 \|U_{i,n}\|^2.$$

Taking conditional expectations and applying the variance bounds $\mathbb{E}[\|U_{i,n}\|^2 \mid \mathcal{F}_n] \leq \sigma_{i,n}^2$ gives the result. □

5 The AM-HM Variance Improvement

This section contains our main new result: a precise quantification of the variance improvement from evidence weighting.

5.1 The Keynesian variance model

Definition 5.1 (Keynesian variance)**.** *Agent i's gradient estimator has variance inversely proportional to its evidence weight:*

$$\sigma_{i,n}^2 = \frac{C}{V_{i,n}}, \tag{16}$$

where $C > 0$ is a constant depending on game parameters (cf. Giannou Lemma 4, $C = 24A^2/\zeta^4$) and $V_{i,n} > 0$ is agent i's Keynesian weight of evidence at episode n.

This model captures the intuition that agents with less evidence produce noisier gradient estimates. It is the natural variance structure when the REINFORCE estimator is computed from observations weighted by evidence quality.

5.2 Unweighted vs. weighted total variance

Under the Keynesian model (16):

Unweighted (standard PG). The total variance is:

$$\sigma_{\text{std}}^2 := \sum_{i=1}^{N} \sigma_{i,n}^2 = C \sum_{i=1}^{N} \frac{1}{V_{i,n}} = \frac{CN}{\text{HM}(V_n)}, \tag{17}$$

where $\text{HM}(V_n) := N/\sum_i V_{i,n}^{-1}$ is the harmonic mean of the evidence weights.

Evidence-weighted (EWPG with $w_{i,n} = V_{i,n}/\bar{V}_n$). Setting $\bar{V}_n = \frac{1}{N}\sum_j V_{j,n} = \text{AM}(V_n)$, the weighted total variance is:

$$\sigma_w^2 := \sum_{i=1}^{N} w_{i,n}^2 \cdot \sigma_{i,n}^2 = \sum_{i=1}^{N} \frac{V_{i,n}^2}{\bar{V}_n^2} \cdot \frac{C}{V_{i,n}} = \frac{C}{\bar{V}_n^2} \sum_{i=1}^{N} V_{i,n} = \frac{CN}{\text{AM}(V_n)}. \tag{18}$$

5.3 The main result

Theorem 5.2 (AM-HM variance improvement)**.** *Under the Keynesian variance model* (16) *and the evidence weight $w_{i,n} = V_{i,n}/\bar{V}_n$, the ratio of weighted to unweighted total variance is:*

$$\boxed{\frac{\sigma_w^2}{\sigma_{\text{std}}^2} = \frac{\text{HM}(V_n)}{\text{AM}(V_n)} \leq 1.} \tag{19}$$

Equality holds if and only if $V_{1,n} = V_{2,n} = \cdots = V_{N,n}$.

Proof. From (17) and (18):

$$\frac{\sigma_w^2}{\sigma_{\text{std}}^2} = \frac{CN/\,\text{AM}(V_n)}{CN/\,\text{HM}(V_n)} = \frac{\text{HM}(V_n)}{\text{AM}(V_n)}.$$

The inequality HM $\leq$ AM is the *AM-HM inequality*, which follows from the Cauchy–Schwarz inequality applied to the vectors $(\sqrt{V_1}, \ldots, \sqrt{V_N})$ and $(1/\sqrt{V_1}, \ldots, 1/\sqrt{V_N})$:

$$N^2 = \left(\sum_{i=1}^{N} 1\right)^2 = \left(\sum_{i=1}^{N} \sqrt{V_i} \cdot \frac{1}{\sqrt{V_i}}\right)^2 \leq \left(\sum_{i=1}^{N} V_i\right)\left(\sum_{i=1}^{N} \frac{1}{V_i}\right),$$

which rearranges to $\text{HM}(V) \leq \text{AM}(V)$. Equality in Cauchy–Schwarz holds iff $\sqrt{V_i}/(1/\sqrt{V_i}) = V_i$ is constant across i. □

Remark 5.3 (Magnitude of improvement)**.** *For $N = 2$ agents with evidence ratio $r = V_1/V_2 \geq 1$:*

$$\frac{\mathrm{HM}}{\mathrm{AM}} = \frac{2r}{(1+r)^2/2} = \frac{4r}{(1+r)^2}.$$

At $r = 1$ (equal evidence): ratio $= 1$ (no improvement). At $r = 10$: ratio $= 40/121 \approx 0.33$ (67% variance reduction). The more heterogeneous the evidence, the larger the gain.

6 Convergence of Evidence-Weighted PG

Theorem 6.1 (Convergence of EWPG to SOS Nash)**.** *Let π^* be an SOS Nash policy with parameter $\mu > 0$. Let $\{\pi_n\}$ be the sequence generated by EWPG* (EWPG) *with:*

(i) Step size $\gamma_n = \gamma/(n+m)^p$, $p \in (1/2, 1]$.

(ii) Evidence weights $w_{i,n} \in [w_{\min}, 1]$ with $w_{\min} > 0$, $\mathcal{F}_n$-measurable.

(iii) Gradient estimates satisfying (4) *with $p + \ell_b > 1$ and $p - \ell_\sigma > 1/2$.*

Then:

(1) There exists a neighbourhood $\mathcal{U}$ of π^* such that $\mathbb{P}(\pi_n \to \pi^* \mid \pi_1 \in \mathcal{U}) \geq 1 - \delta$ for any $\delta > 0$, provided γ is small enough.

(2) The convergence rate is:

$$\mathbb{E}\big[\|\pi_n - \pi^*\|^2 \mid \mathcal{E}\big] = \mathcal{O}\left(\frac{C_w}{n^q}\right), \tag{20}$$

where $q = \min\{\ell_b, p - 2\ell_\sigma\}$ and the constant C_w satisfies:

$$\frac{C_w}{C_{\mathrm{std}}} = \frac{\mathrm{HM}(V)}{\mathrm{AM}(V)} \leq 1, \tag{21}$$

with C_{std} being the constant from Giannou et al. Theorem 2.

Proof sketch. The proof follows the structure of Giannou et al. Appendices A–B, with the substitution $\gamma_n \to \gamma_n w_{i,n}$ for each agent i. We trace where the variance enters the bounds.

Step 1: Modified energy inequality. By Theorem 4.1, the Lyapunov function satisfies (10). Using the SOS condition (2):

$$\langle W_n v(\pi_n), \pi_n - \pi^* \rangle \leq -\mu \sum_{i=1}^{N} w_{i,n} \|\pi_{i,n} - \pi_i^*\|^2 \leq -\mu w_{\min} \|\pi_n - \pi^*\|^2. \tag{22}$$

This gives a contraction rate of $2\mu w_{\min}\gamma_n$ (compared to $2\mu\gamma_n$ in the standard analysis). The effective SOS parameter becomes $\mu_w := \mu \cdot w_{\min}$.

Step 2: Error control. Following Giannou's Appendix A.2, we bound the cumulative error. The noise term ξ_n^w has the same martingale property ($\mathbb{E}[\xi_n^w \mid \mathcal{F}_n] = 0$) and the variance bound:

$$\mathbb{E}[(\xi_n^w)^2 \mid \mathcal{F}_n] \leq \|\pi_n - \pi^*\|^2 \cdot \sum_{i=1}^{N} w_{i,n}^2 \sigma_{i,n}^2 = \|\pi_n - \pi^*\|^2 \cdot \sigma_w^2.$$

By Theorem 5.2, $\sigma_w^2 = (\mathrm{HM} \,/\, \mathrm{AM}) \cdot \sigma_{\mathrm{std}}^2 \leq \sigma_{\mathrm{std}}^2$.

Step 3: Convergence rate. Following Giannou's Appendix B, the conditional Lyapunov bound becomes (cf. their eq. B.8):

$$\bar{D}_{n+1} \leq \left(1 - \frac{2\mu_w \gamma}{(n+m)^p}\right) \bar{D}_n + \frac{C_1}{(n+m)^{p+\ell_b}} + \frac{C_2^w}{(n+m)^{2p-2\ell_\sigma}}, \tag{23}$$

where $C_2^w = (\mathrm{HM} \,/\, \mathrm{AM}) \cdot C_2^{\mathrm{std}}$. The rest of the Chung lemma argument proceeds identically, yielding the rate $\mathcal{O}(C_w/n^q)$ with the improved constant. □

Corollary 6.2 (REINFORCE with evidence weighting)**.** *For ε-greedy REINFORCE (Giannou Algorithm 2) with evidence-weighted updates:*

(a) With $\gamma_n = \gamma/(n+m)$, $\varepsilon_n = \varepsilon/(n+m)^{1/2}$, the convergence rate is $\mathcal{O}(C_w/\sqrt{n})$.

(b) The constant satisfies $C_w/C_{\text{std}} = \text{HM}(V)/\text{AM}(V) \leq 1$.

(c) For $N = 2$ agents with evidence ratio $r = V_1/V_2 = 10$, the leading constant is reduced by approximately 67%.

7 Optimal Evidence Weight

Proposition 7.1 (Optimal evidence weight)**.** *Under the Keynesian variance model $\sigma^2_{i,n} = C/V_{i,n}$, the evidence weight that minimises the weighted total variance $\sigma^2_w = \sum_i w^2_{i,n}\sigma^2_{i,n}$ subject to the drift constraint $\sum_i w_{i,n}g_{i,n} = \sum_i g_{i,n}$ (where $g_{i,n} = \langle v_i(\pi_n), \pi_{i,n} - \pi^*_i\rangle$) is:*

$$w^*_{i,n} \propto V_{i,n} \cdot g_{i,n}. \tag{24}$$

When the per-agent drift terms $g_{i,n}$ are approximately equal, this simplifies to $w^*_{i,n} \propto V_{i,n}$, i.e. $w_{i,n} = V_{i,n}/\bar{V}_n$.

Proof. Lagrangian: $L = \sum_i w_i^2 C/V_i - \lambda(\sum_i w_i g_i - \sum_i g_i)$. First-order condition:

$$\frac{\partial L}{\partial w_i} = \frac{2w_i C}{V_i} - \lambda g_i = 0 \implies w_i = \frac{\lambda V_i g_i}{2C}.$$

The constraint $\sum_i w_i g_i = \sum_i g_i$ determines λ. When $g_i \approx g$ for all i:

$$w_i = \frac{\lambda V_i g}{2C}, \quad \sum_i \frac{\lambda V_i g^2}{2C} = Ng \implies \lambda = \frac{2CN}{g\sum_i V_i},$$

yielding $w_i = V_i N/\sum_j V_j = V_i/\bar{V}$. □

8 Connection to the Ω-Framework

The evidence-weighted PG method instantiates several components of the Ω-framework multi-agent gradient (theos eq. 27):

$$\nabla^{\text{full}}_{\theta_i} R_{\text{multi}} = \omega_i \Big[\underbrace{\mathbb{E}[\mathcal{L}_i \cdot s_{\theta_i}]}_{\text{exploration}} + \underbrace{\mathbb{E}[\nabla_{\theta_i}\mathcal{L}_i]}_{\text{exploitation}}\Big] + \omega_i\mu \underbrace{(-V_i^{-2}\nabla_{\theta_i}V_i)}_{\text{evidence seeking}} + \lambda \underbrace{\nabla_{\theta_i} D}_{\text{alignment}} + \underbrace{\sum_{j\neq i} \frac{\partial R_i}{\partial \theta_j} \cdot \frac{\partial \theta'_j}{\partial \theta_i}}_{\text{opponent shaping}}. \tag{25}$$

Our evidence-weighted PG captures the first three components:

- The **exploration** and **exploitation** terms are the standard REINFORCE gradient $\hat{v}_{i,n}$.
- The **evidence seeking** term is operationalised through the evidence weight $w_{i,n}$: by scaling the step size by V_i, agents implicitly optimise the Keynesian loss that trades prediction quality against evidentiary robustness.

The **alignment** and **opponent shaping** terms are not included in the current analysis. Extending Theorem 6.1 to incorporate LOLA-type opponent shaping (cf. Foerster et al. 2018) is a natural direction for future work.

Gödelian complementarity made quantitative. The AM-HM improvement factor $\mathrm{HM}(V)/\mathrm{AM}(V)$ is a quantitative expression of Gödelian complementarity (theos item 37): agents with diverse evidence quality (heterogeneous V_i) collectively achieve a tighter variance bound than homogeneous agents, not by being individually better, but by *having different strengths*. The harmonic mean rewards the collective for the diversity of its perspectives.

9 Discussion

What is new. The contribution is the observation that Keynesian evidence weights from the Ω-framework, when used as adaptive step-size multipliers in multi-agent PG, produce a provable variance improvement of $\mathrm{HM}(V)/\mathrm{AM}(V)$. The proof is a modification of Giannou et al.'s Lyapunov analysis, not a new technique, but the result connects two previously separate literatures: convergence theory for MARL and the epistemology of incomplete knowledge.

What is not new. The convergence rate exponent q is unchanged. The improvement is in the constant, which matters in practice (it determines how many episodes are needed to reach a given accuracy) but does not change the asymptotic scaling.

Limitations.

1. The Keynesian variance model $\sigma_i^2 = C/V_i$ is assumed, not derived from first principles. In practice, the relationship between evidence quality and gradient variance depends on the specific estimator.
2. The evidence weights $V_{i,n}$ must be observable or estimable by the agents. In some settings this is natural (e.g. sample counts); in others it requires additional machinery.
3. The improvement vanishes when all agents have equal evidence. The result is most useful precisely when agents face heterogeneous data environments.

Future directions.

1. Extend to LOLA/opponent-shaping updates (Candidate 2 from the project notes).
2. Prove a basin-of-attraction enlargement result under agent diversity (Candidate 3).
3. Implement EWPG in the existing simulation framework and compare empirically to standard PG on the matching pennies and iterated prisoner's dilemma environments.

References

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A Pseudocode

Algorithm 1: Evidence-Weighted Policy Gradient (EW-PG)

Input: Stochastic game $\mathcal{G}$, N agents, learning rates $\gamma_n = \gamma/(n+m)^p$
Input: Evidence decay $\alpha \in (0,1)$, minimum evidence $V_{\min} > 0$
Output: Policies $\{\pi_i\}_{i=1}^N$ converging to stable Nash

Initialize $\pi_i \in \Pi$ for each agent i;
Initialize running variance estimates $\hat{V}_i \leftarrow V_{\min}$ for each i;
for $n = 1, 2, \ldots$ **do**
  `// --- Phase 1: Rollout and gradient estimation ---`
  for $i = 1, \ldots, N$ **do**
    Collect trajectory $\tau_i \sim \pi_i$ in current game state;
    Compute policy gradient estimate $\hat{v}_{i,n}$ via REINFORCE:;
      $\hat{v}_{i,n} \leftarrow \frac{1}{|\mathcal{B}|} \sum_{\tau \in \mathcal{B}} R(\tau) \cdot \nabla_{\theta_i} \log \pi_{\theta_i}(\tau)$;
  `// --- Phase 2: Evidence estimation ---`
  for $i = 1, \ldots, N$ **do**
    Update variance estimate (exponential moving average):;
    $\hat{V}_{i,n} \leftarrow \alpha \cdot \hat{V}_{i,n-1} + (1-\alpha) \cdot \|\hat{v}_{i,n} - \bar{v}_i\|^2$;
    Clamp: $\hat{V}_{i,n} \leftarrow \max(\hat{V}_{i,n},\ V_{\min})$;
  `// --- Phase 3: Evidence-weighted update ---`
  $V_{\min,n} \leftarrow \min_i \hat{V}_{i,n}$;
  for $i = 1, \ldots, N$ **do**
    Compute Keynesian evidence weight:;
    $w_{i,n} \leftarrow V_{\min,n} \,/\, \hat{V}_{i,n} \qquad \in (0,1]$;
    Update policy with evidence-weighted step:;
    $\pi_{i,n+1} \leftarrow \mathrm{proj}_{\Pi}\big(\pi_{i,n} + \gamma_n \cdot w_{i,n} \cdot \hat{v}_{i,n}\big)$;

Remark A.1 (Implementation notes). *1. The variance estimate $\hat{V}_{i,n}$ uses exponential moving average for stability. Alternative: batch variance over a sliding window.*

2. The minimum evidence $V_{\min}$ prevents division by zero and ensures all agents make some progress. In practice, set $V_{\min} = 10^{-6}$.

3. The projection proj_{Π} is onto the simplex $\Delta(\mathcal{A}_i)^{|\mathcal{S}|}$, implementable in $O(|\mathcal{A}| \log |\mathcal{A}|)$ via sorting.

4. The variance improvement ratio $\mathrm{HM}(V)/\,\mathrm{AM}(V)$ is not computed explicitly — it emerges from the weighted updates. It can be monitored as a diagnostic: $\rho_n = N \cdot V_{\min,n} / \sum_i \hat{V}_{i,n}$.